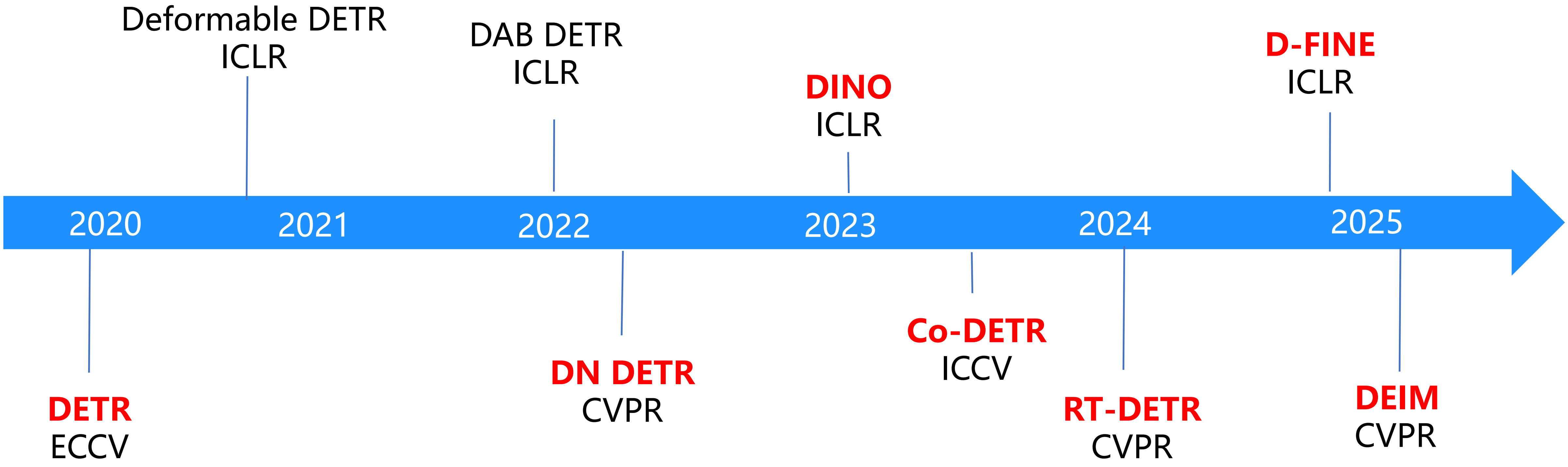
基于Transformer的目标检测

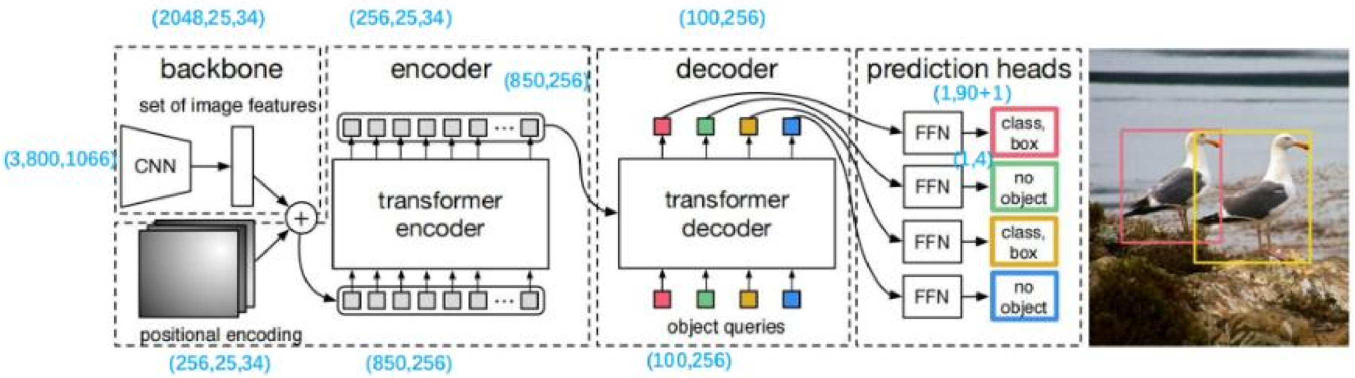
从DETR到Co-DETR、RT-DETR、D-FINE、DEIM

原理+代码

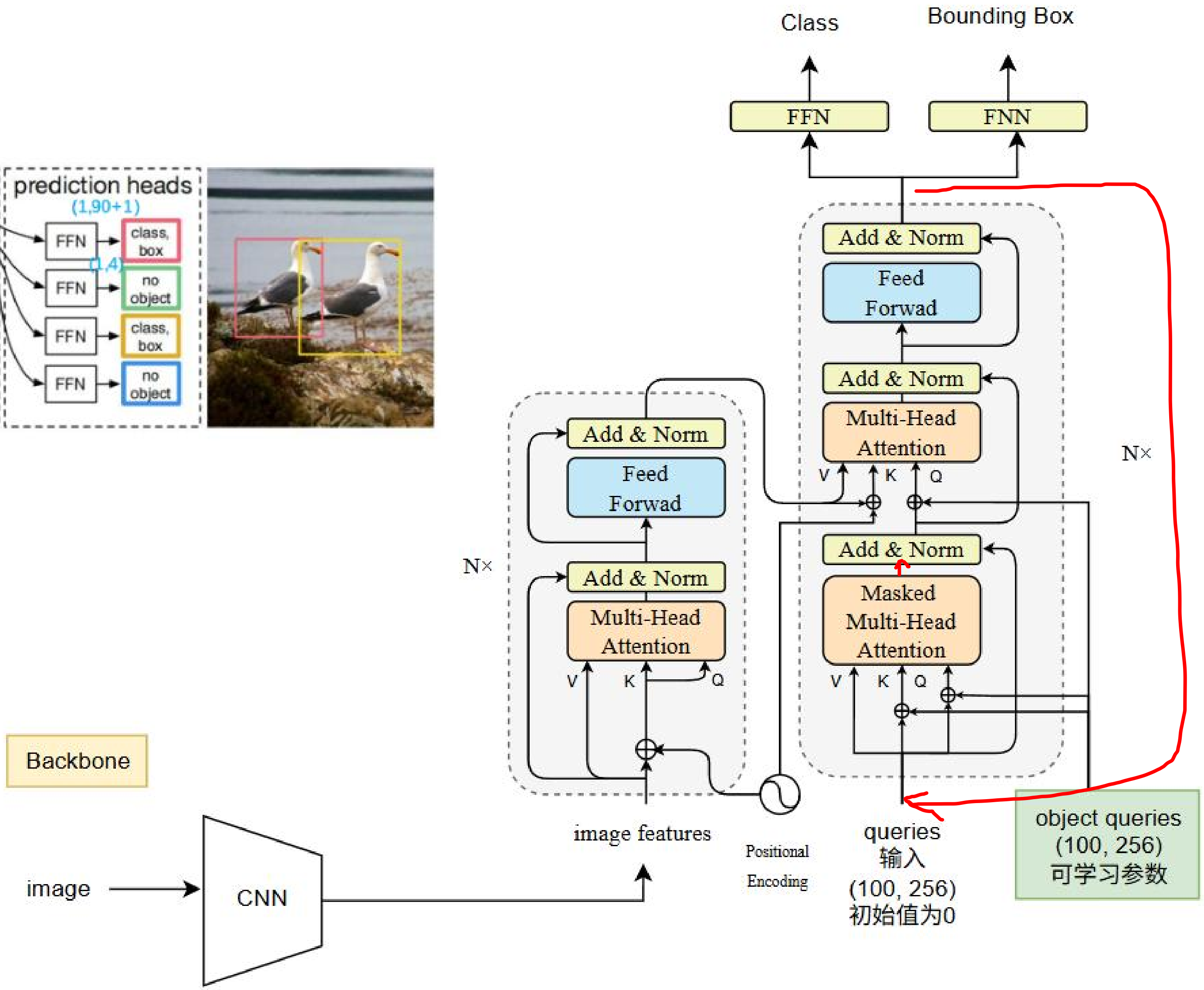
DETR系列进程


Deformable DETR
ICLR
DAB DETR
ICLR
DINO
ICLR
D-FINE
ICLR
2020
2021
2022
2023
2024
2025
DETR
ECCV
DN DETR
CVPR
Co-DETR
ICCV
RT-DETR
CVPR
DEIM
CVPR

DETR

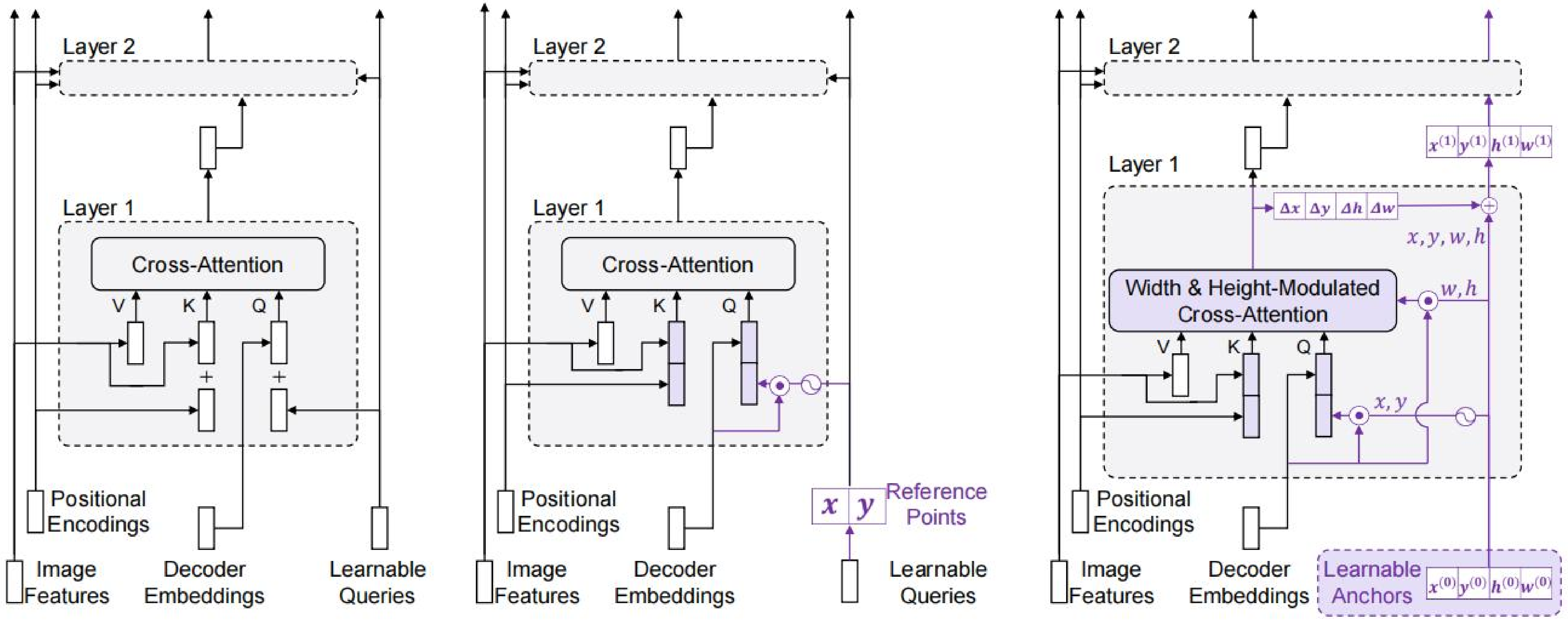


(2048,25,34)
(256,25,34)
(100,256)
backbone
set of image features
encoder
(850,256)
decoder
prediction heads
(1,90+1)
(3,800,1066)
CNN
transformer encoder
transformer decoder
FFN
(1,4)
class, box
no object
positional encoding
object queries
(256,25,34)
(850,256)
(100,256)

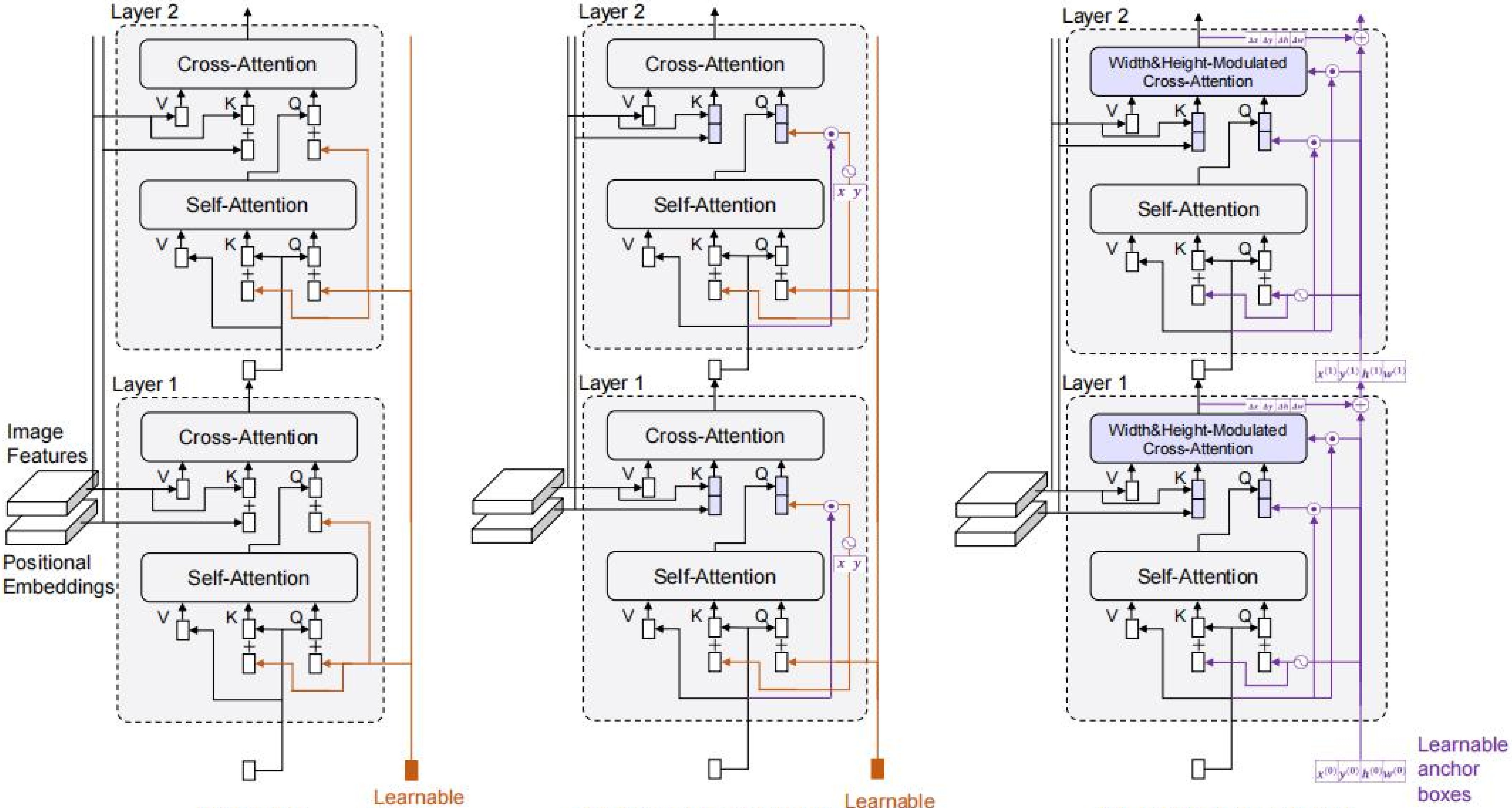


Class
Bounding Box
FFN
FNN
Add & Norm
Feed Forwad
Multi-Head Attention
Masked Multi-Head Attention
V
K
Q
N×
Backbone
image
CNN
image features
Positional Encoding
queries
输入
(100, 256)
初始值为0
object queries
(100, 256)
可学习参数

DAB-DETR

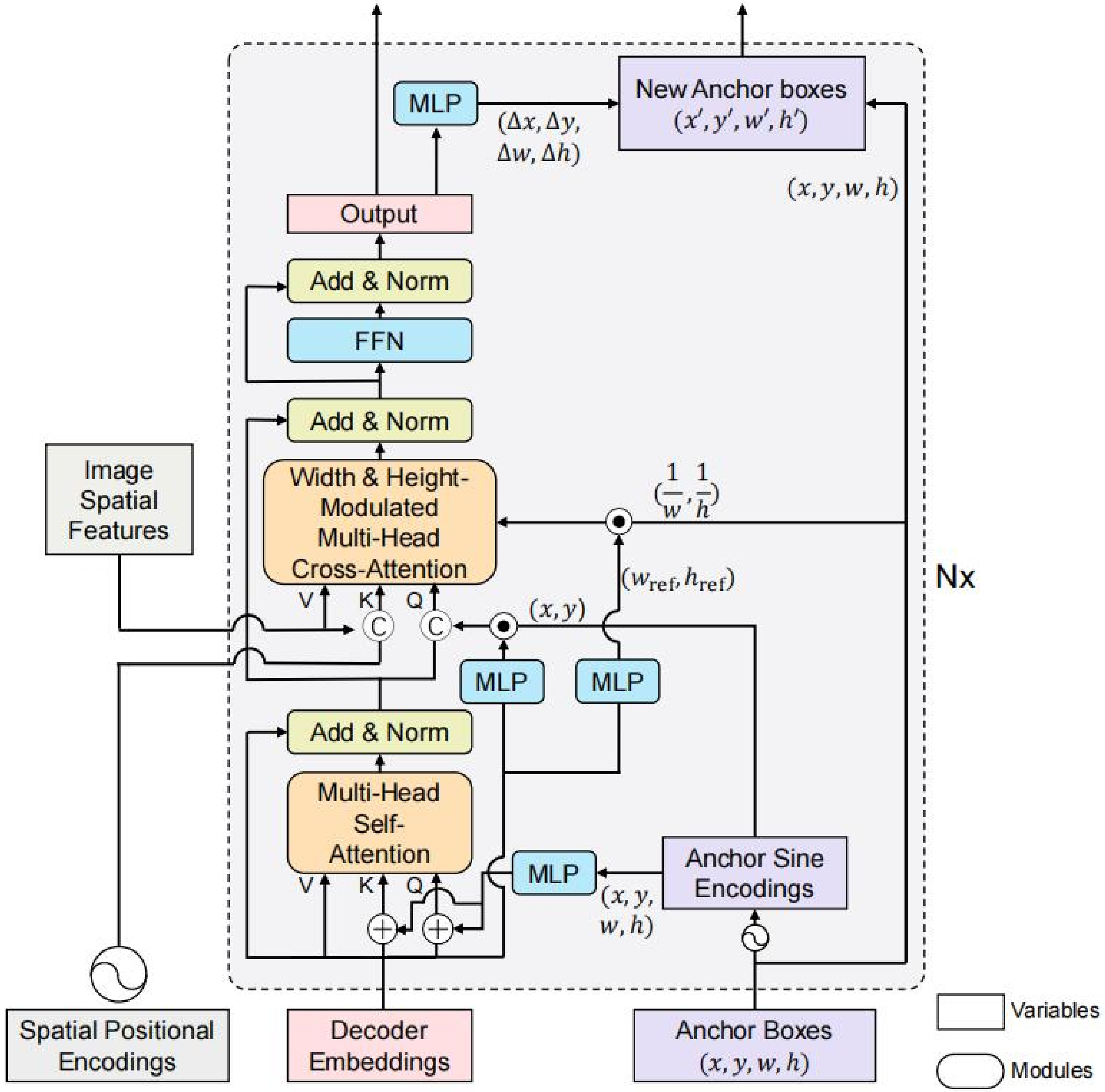


(a) DETR (b) Conditional DETR (c) DAB-DETR



(a) DETR (b) Conditional DETR (c) DAB-DETR (Ours)

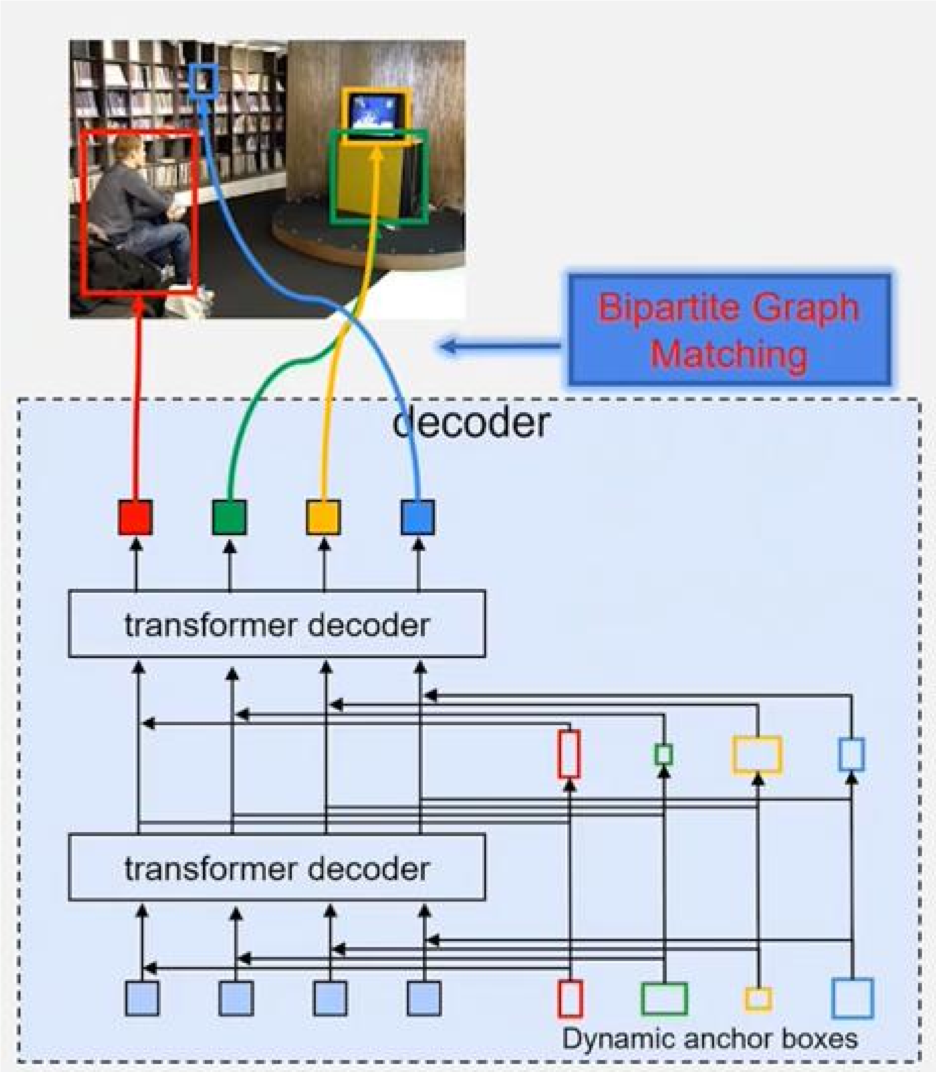
DAB-DETR

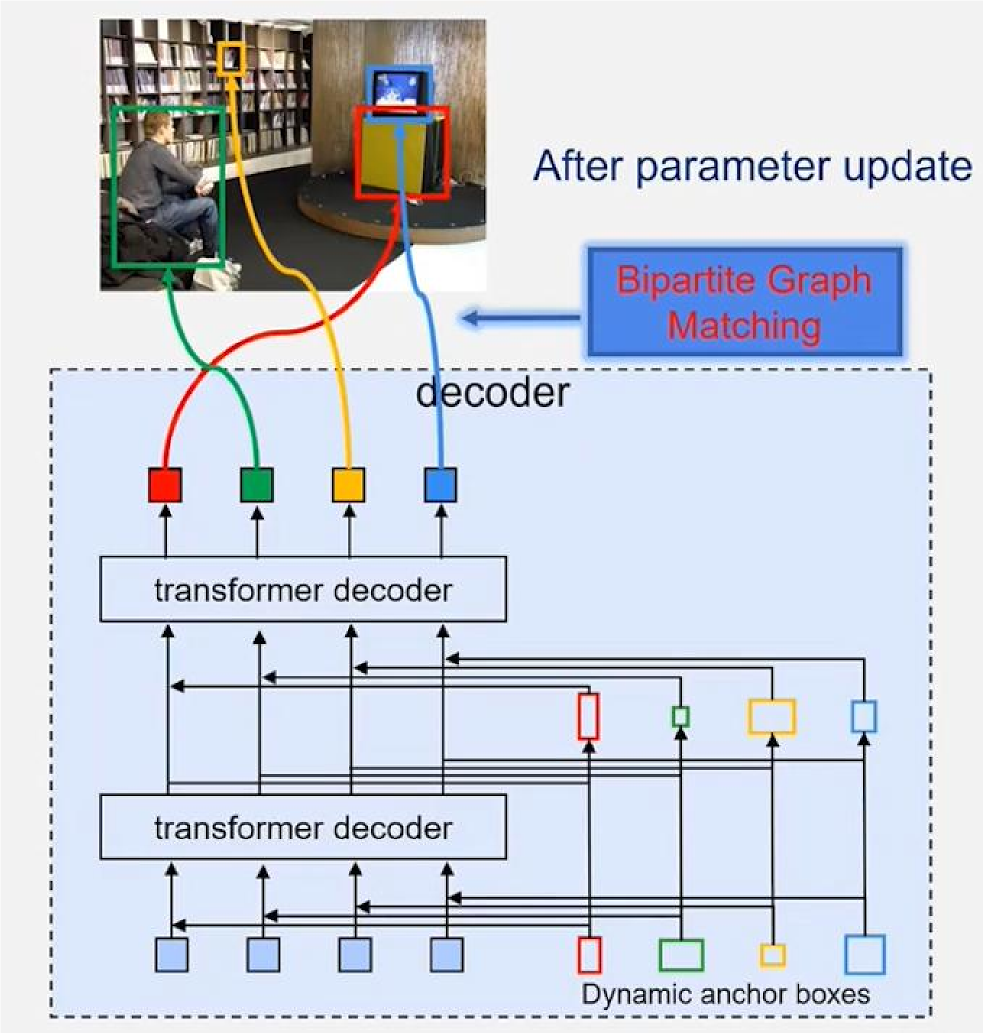


New Anchor boxes
(x', y', w', h')
MLP
$(\Delta x, \Delta y, \Delta w, \Delta h)$
(x, y, w, h)
Output
Add & Norm
FFN
Add & Norm
Image Spatial Features
Width & Height-Modulated Multi-Head Cross-Attention
$(\frac{1}{w}, \frac{1}{h})$
(w_{ref}, h_{ref})
Nx
V K Q
C C
(x, y)
MLP
MLP
Add & Norm
Multi-Head Self-Attention
V K Q
MLP
Anchor Sine Encodings
(x, y, w, h)
Spatial Positional Encodings
Decoder Embeddings
Anchor Boxes
(x, y, w, h)
Variables
Modules

DN-DETR

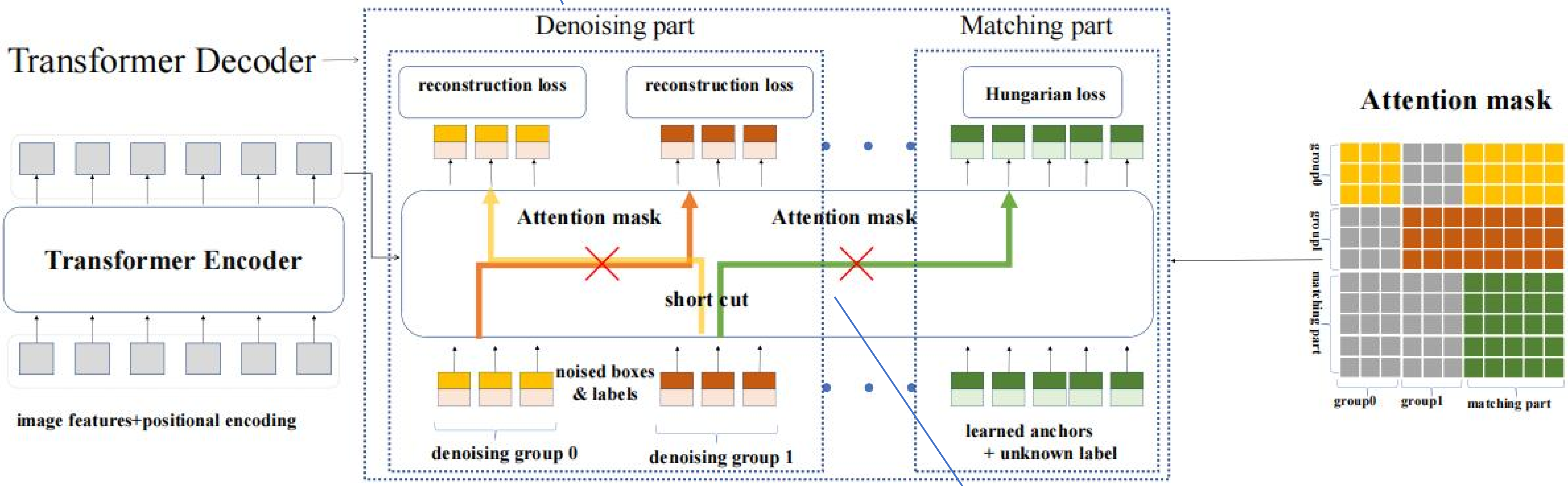
二分图匹配选择不稳定





DN-DETR

加噪的Objects采用重构loss，任务简单

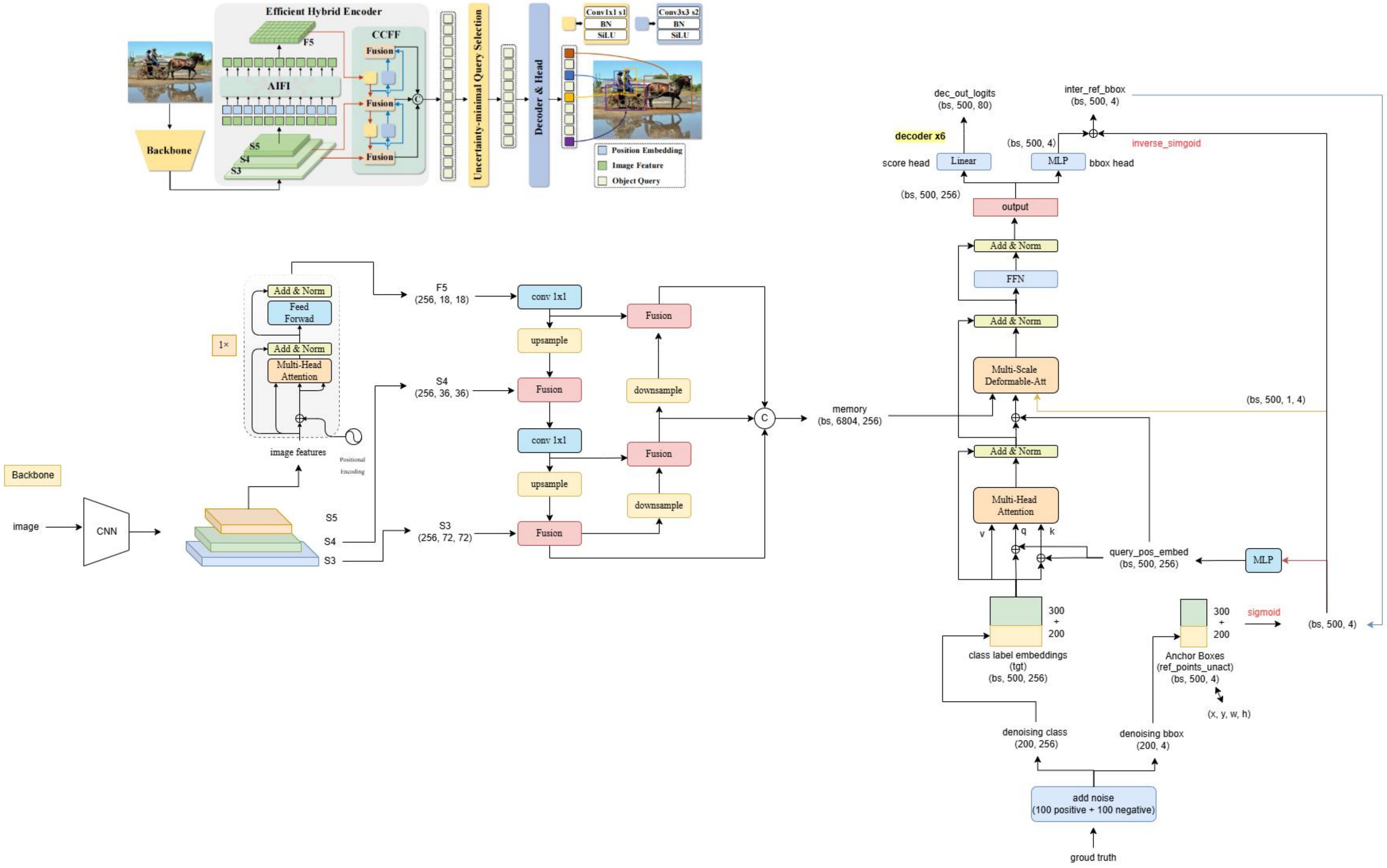


加噪的GT objects

Attention Mask防止信息泄露

RT-DETR

AIFI: Attention-based Intra-scale feature interaction
CCFF: Cross-scale Feature Fusion



Co-DETR为什么能成为SOTA?

传统目标检测算法和新兴端到端目标检测算法的集大成者

强有力的骨干网络 + 多头辅助训练

一对一：一个GT对应一个正样本

只选最匹配正样本的计算损失

一对多：一个GT对应多个正样本

选择多个较为匹配的正样本计算损失

Co-DETR

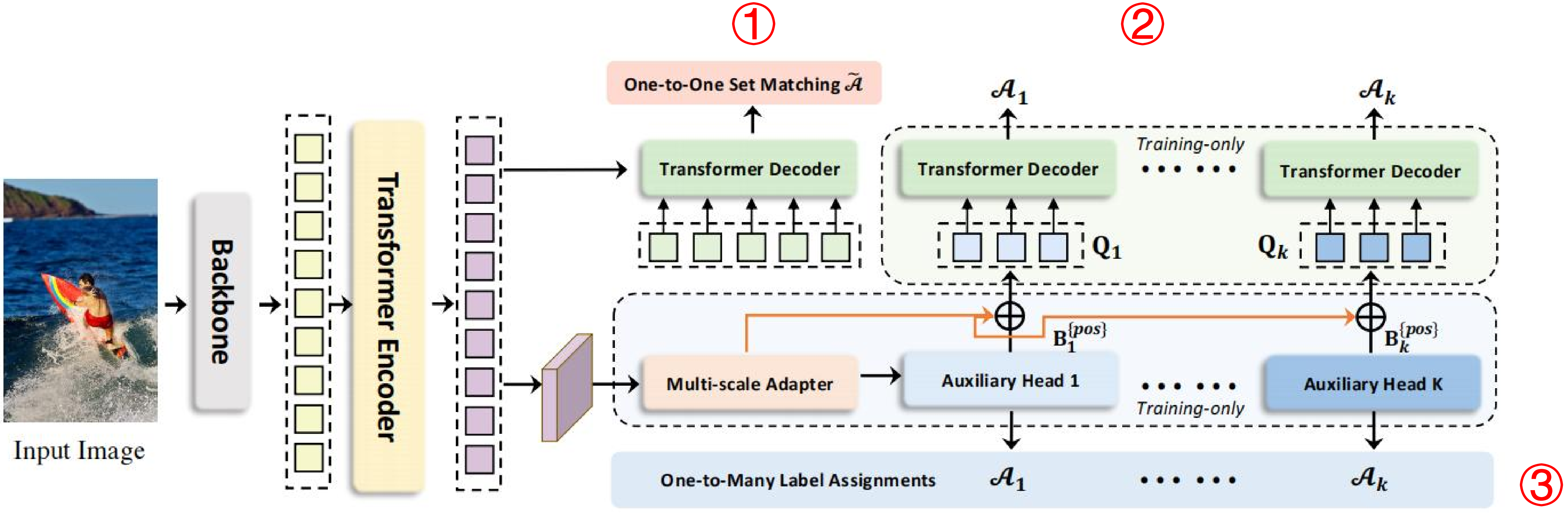


Figure 4. **Framework of our Collaborative Hybrid Assignment Training.** The auxiliary branches are discarded during evaluation.

$$\mathcal{L}^{global} = \sum_{l=1}^{L} (\widetilde{\mathcal{L}}_l^{dec} + \lambda_1 \sum_{i=1}^{K} \mathcal{L}_{i,l}^{dec} + \lambda_2 \mathcal{L}^{enc}), \quad (6)$$

① ② ③

https://zhuanlan.zhihu.com/p/715082004

DETR

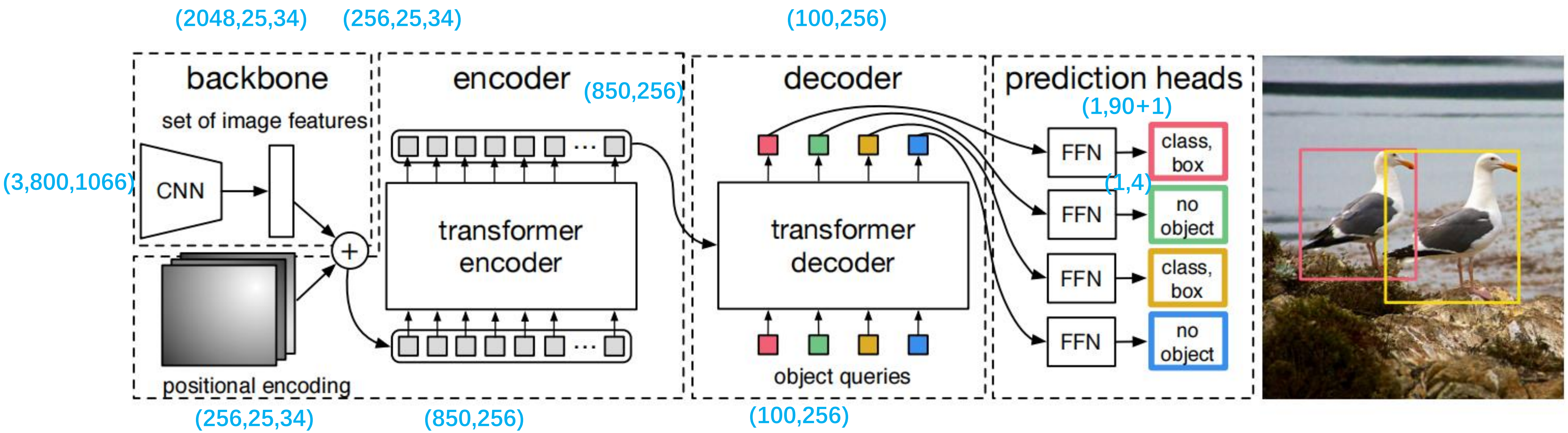
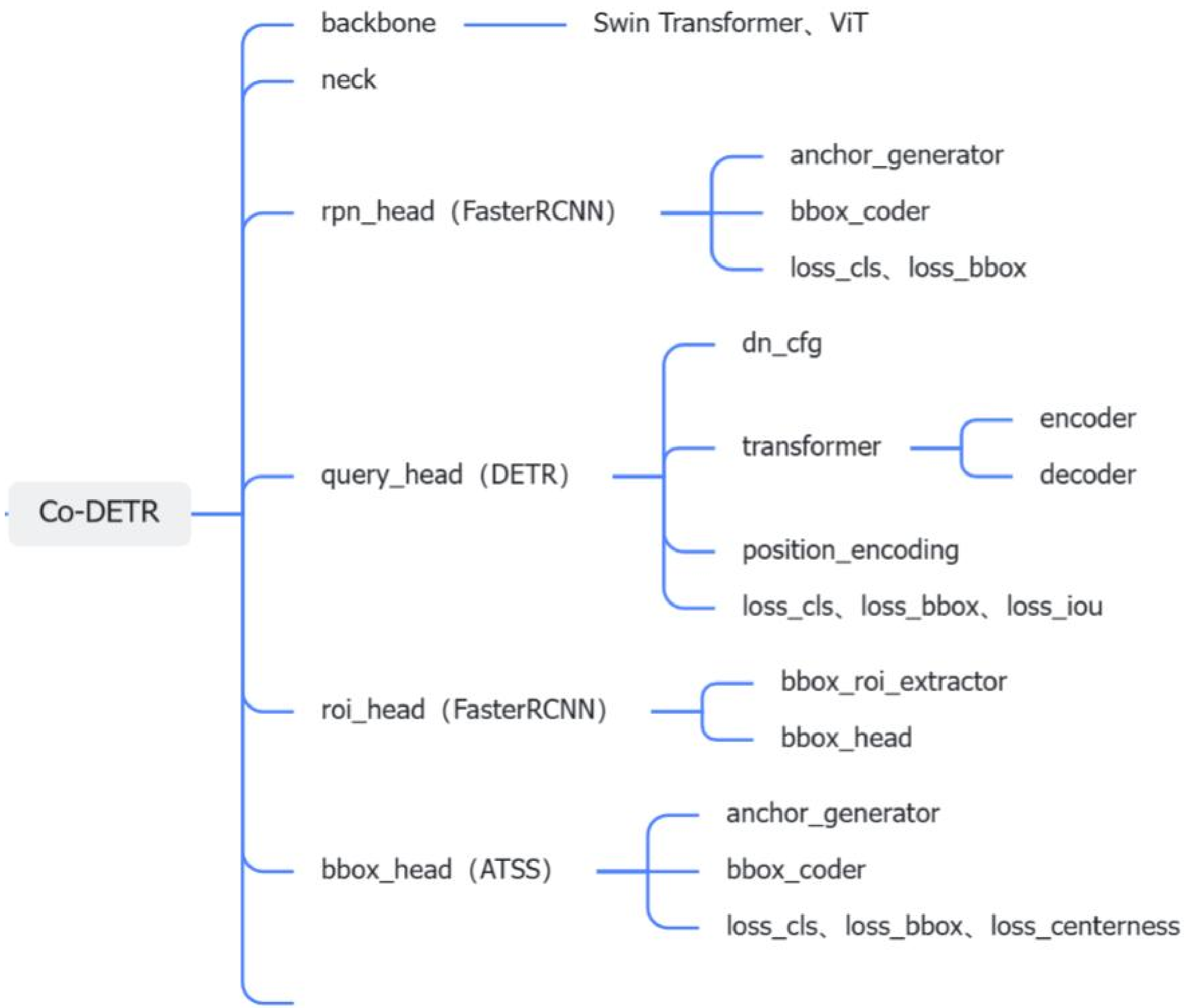
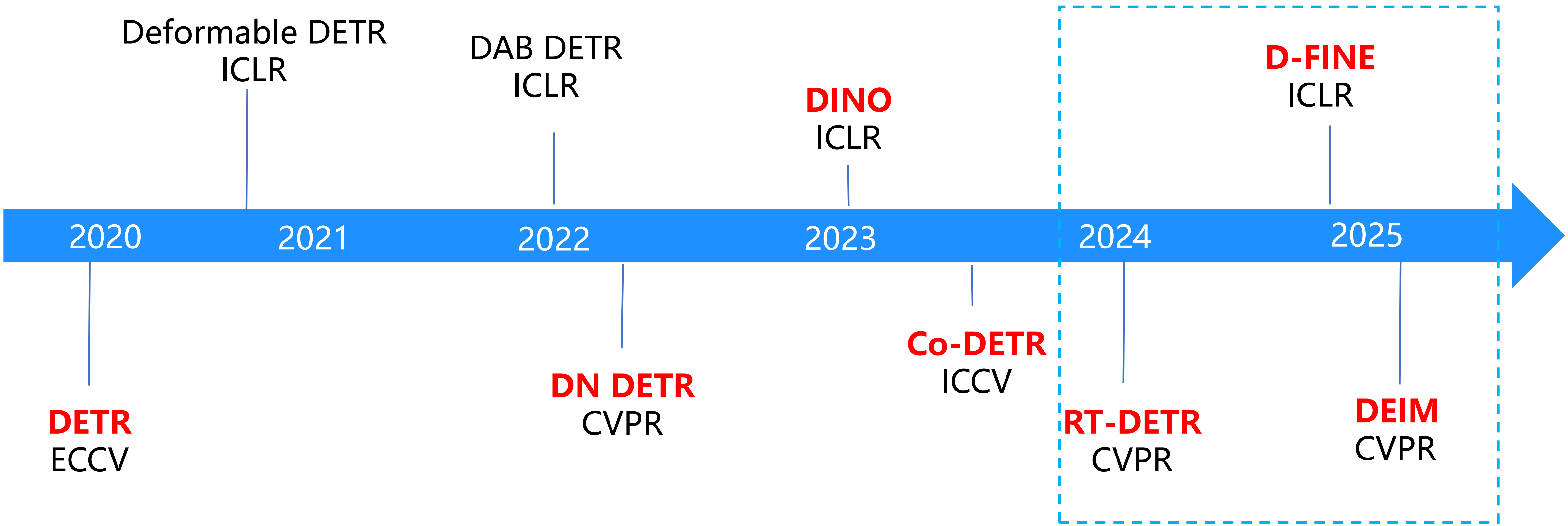


图2：DETR 使用传统的 CNN 主干网络来学习输入图像的 2D 表示。模型将该表示展平，并补充位置编码，然后将其传递给Transformer编码器。Transformer解码器以少量固定数量的学习位置嵌入（称之为对象查询）作为输入，并且还关注编码器的输出。将解码器的每个输出嵌入传递给一个共享的前馈网络（FFN），该网络预测一个检测（类别和边界框）或“无对象”类别。

配置文件


Co-DETR
backbone
Swin Transformer、ViT
neck
rpn_head（FasterRCNN）
anchor_generator
bbox_coder
loss_cls、loss_bbox
query_head（DETR）
dn_cfg
transformer
encoder
decoder
position_encoding
loss_cls、loss_bbox、loss_iou
roi_head（FasterRCNN）
bbox_roi_extractor
bbox_head
bbox_head（ATSS）
anchor_generator
bbox_coder
loss_cls、loss_bbox、loss_centerness

DEIM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DEIM

D-FINE: REDEFINE REGRESSION TASK IN DETRS AS FINE-GRAINED DISTRIBUTION REFINEMENT

Table 3: Step-by-step modifications from baseline model to D-FINE. Each step shows changes in AP, the number of parameters, latency, and FLOPs.

RT-DETR

Model	AP^{val}	#Params.	Latency (ms)	GFLOPs
baseline: RT-DETR-HGNetv2-L (Zhao et al., 2024)	53.0	32M	9.25	110
Remove Decoder Projection Layers	52.4	32M	8.02	97
+ **Target Gating Layers**	52.8	33M	8.15	98
Encoder CSP layers → GELAN (Wang & Liao, 2024)	53.5	46M	10.69	167
Reduce Hidden Dimension in GELAN by half	52.8	31M	8.01	91
Uneven Sampling Points (S: 3, M: 6, L: 3)	52.9	31M	7.90	91
RT-DETRv2 Training Strategy (Lv et al., 2024)	53.0	31M	7.90	91
+ **FDR**	53.5	31M	8.07	91
+ **GO-LSD**	**54.0** (+1.0)	**31M** (-3%)	**8.07**(-13%)	**91** (-17%)

D-FINE

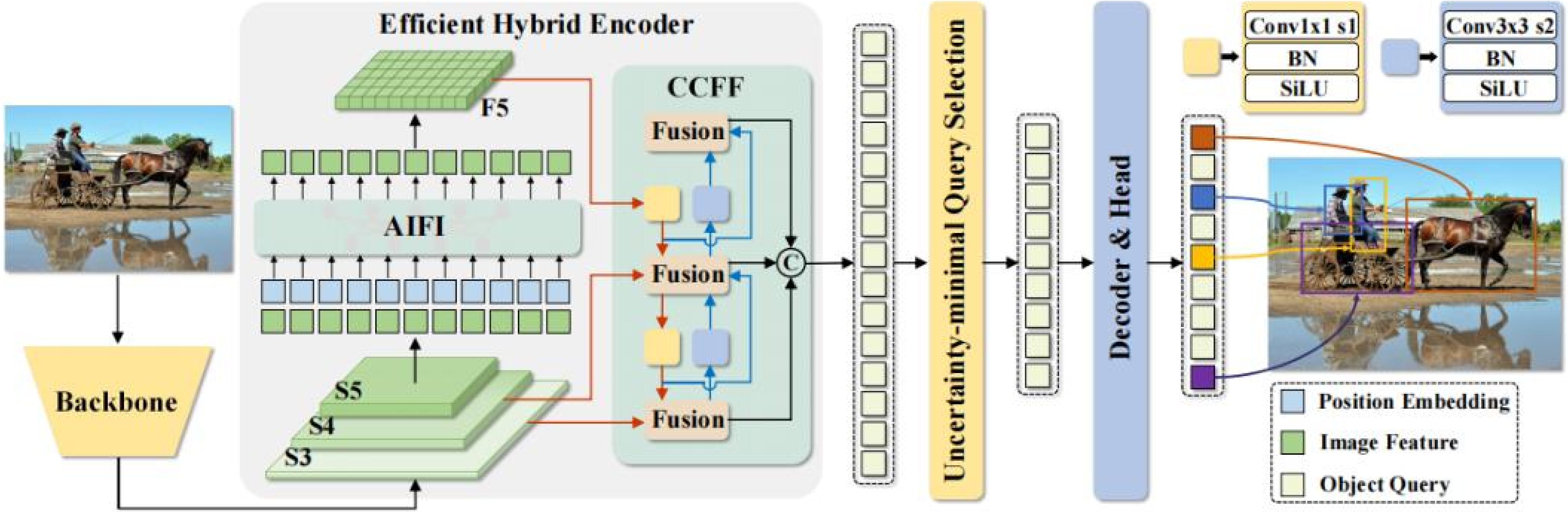
DEIM: DETR with Improved Matching for Fast Convergence

Table 6. **Impact of Dense O2O and MAL.** We conduct experiments with RT-DETRv2-R50 [24] and D-FINE-L [27].

Epochs	Dense O2O	MAL	AP	AP_{50}	AP_{75}
RT-DETRv2-R50 [24]					
72			53.4	71.6	57.4
36	✓		53.6	71.9	58.2
	✓	✓	**53.9**	**71.7**	**58.6**
D-FINE-L [27]					
72			54.0	71.6	58.4
36	✓		54.2	72.1	58.9
	✓	✓	**54.6**	**72.2**	**59.5**

DEIM

RT-DETR

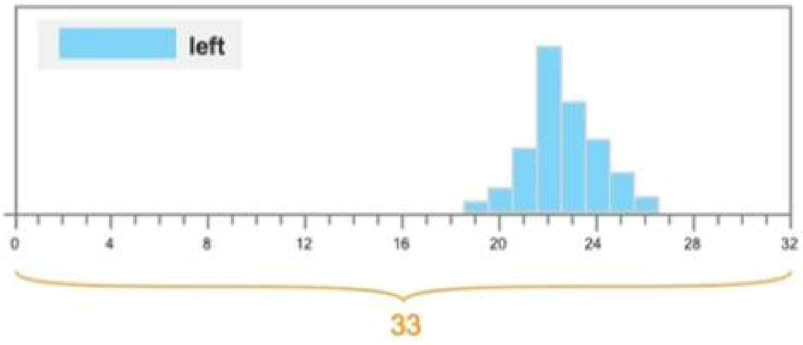


Efficient Hybrid Encoder
F5
CCFF
Fusion
Fusion
Fusion
AIFI
S5
S4
S3
Backbone
Uncertainty-minimal Query Selection
Decoder & Head
Conv1x1 s1
BN
SiLU
Conv3x3 s2
BN
SiLU
Position Embedding
Image Feature
Object Query

D-FINE（FDR）

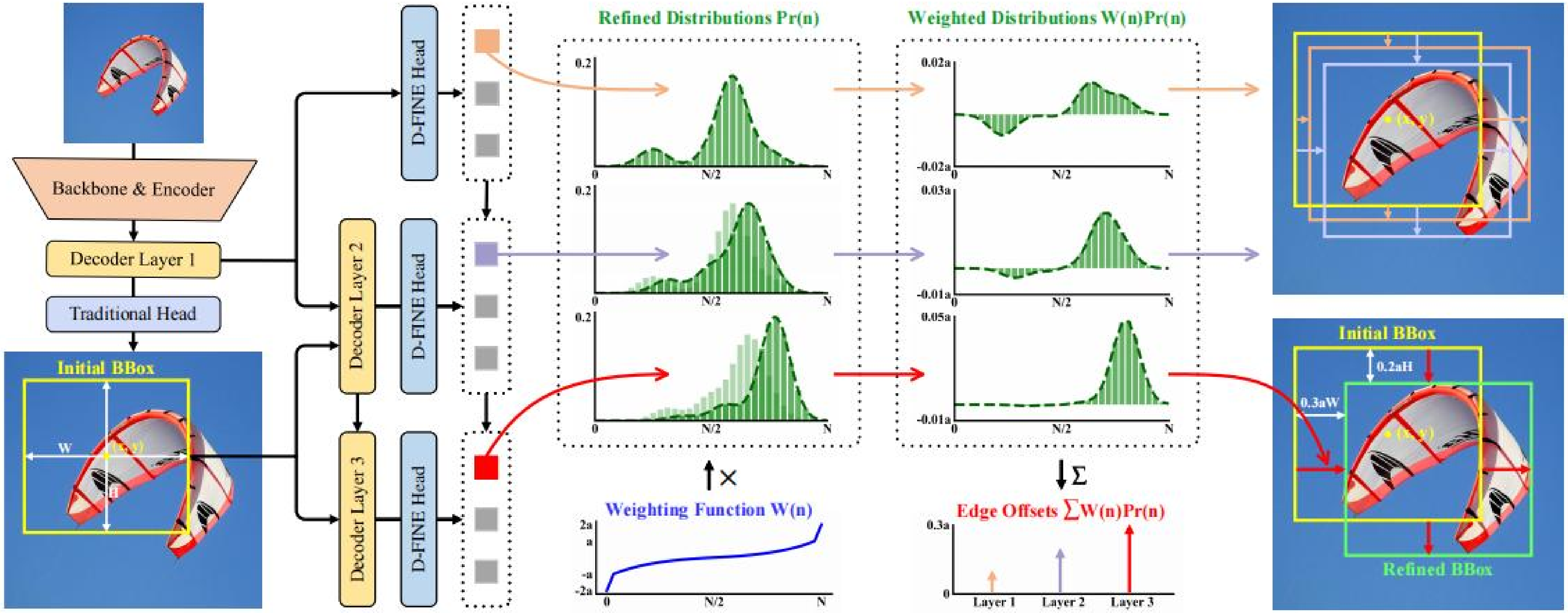
细粒度分布优化，Fine-grained Distribution Refinement (FDR)

bounding box	-->	bounding box distribution
(x, y, w, h)	-->	$(\vec{x}, \vec{y}, \vec{w}, \vec{h})$
(l, r, b, t)	-->	$(\vec{l}, \vec{r}, \vec{b}, \vec{t})$
1 x 4	-->	33 x 4



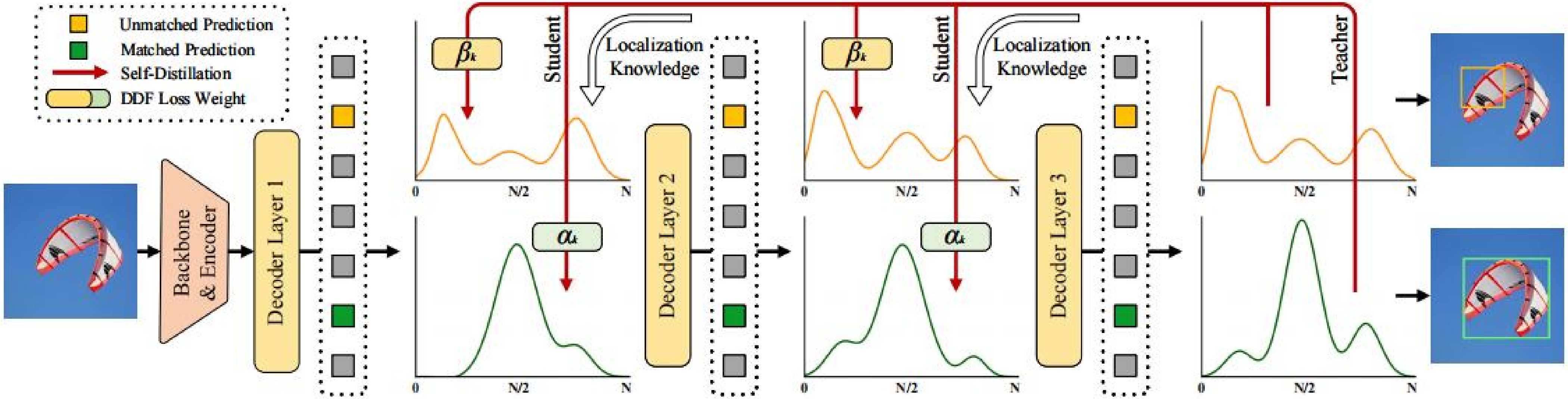
D-FINE（FDR）

FDR 迭代优化作为边界框预测修正的概率分布，提供更细粒度的中间表示。该方法独立捕捉和优化每个边界的不确定性。通过利用非均匀加权函数，FDR 允许在每个解码器层进行更精确的增量调整，提高定位精度并减少预测误差。FDR 在无锚端到端框架内运行，实现更灵活鲁棒的优化过程。



D-FINE（GO-LSD）

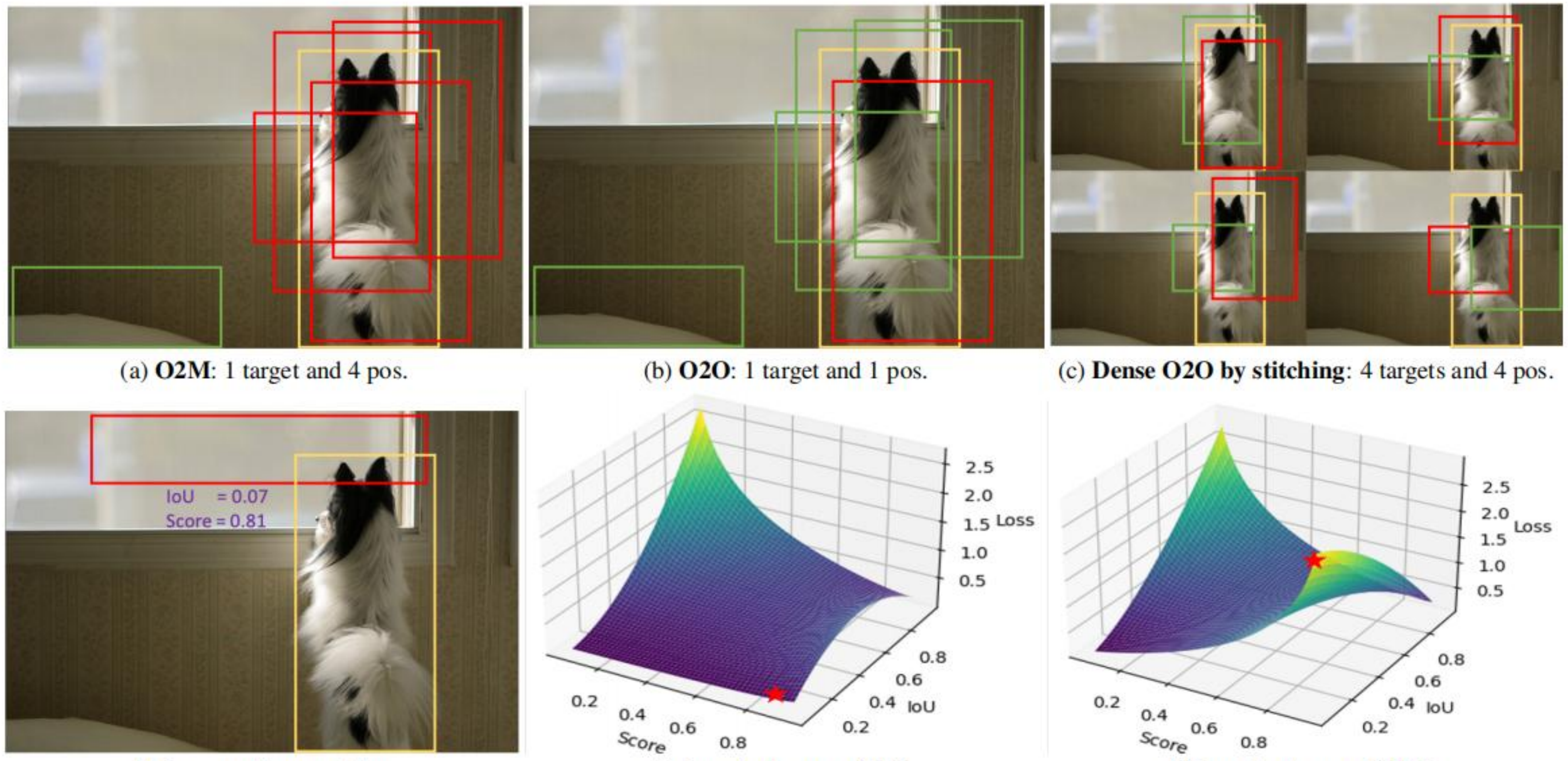
全局最优定位自蒸馏，Global Optimal Localization Self-Distillation (GO-LSD)



DEIM（Dense O2O & MAL）

黄色、红色、绿色框分别表示GT、positive和negative。

图 c 所示，将原始图像复制到四个象限并组合成单张合成图像，保持原始图像尺寸，使目标数从 1 增加到 4，在不改变匹配结构的情况下提升监督水平。Dense O2O 实现了与 O2M 相当的监督水平。



(a) **O2M**: 1 target and 4 pos. (b) **O2O**: 1 target and 1 pos. (c) **Dense O2O by stitching**: 4 targets and 4 pos.

(d) Low-quality matching (e) Loss landscape of VFL (f) Loss landscape of MAL

谢谢观看